

Some Assembly Required:

Simplified Damage Functions for Probabilistic Coastal Impact Estimation Using FrEDI

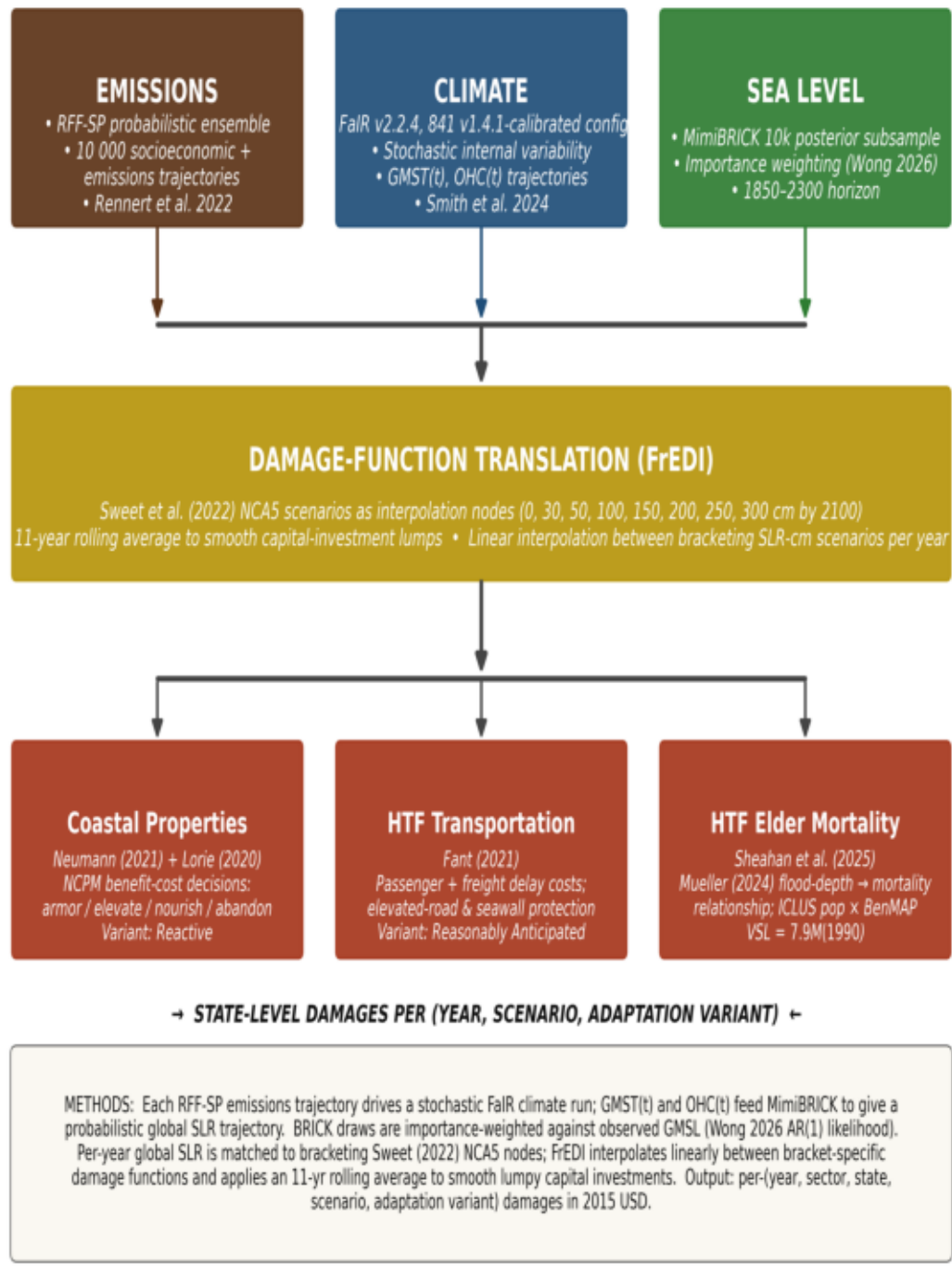
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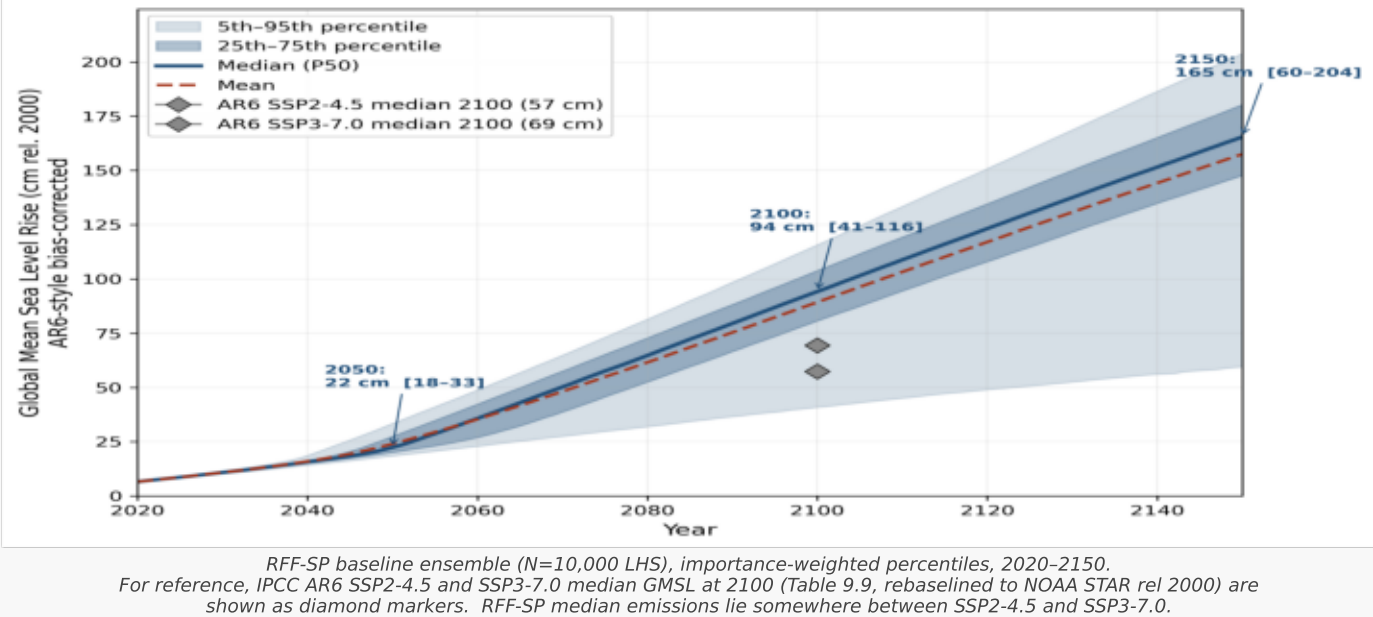
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A. PIPELINE



B. PROBABILISTIC SLR



DISCUSSION

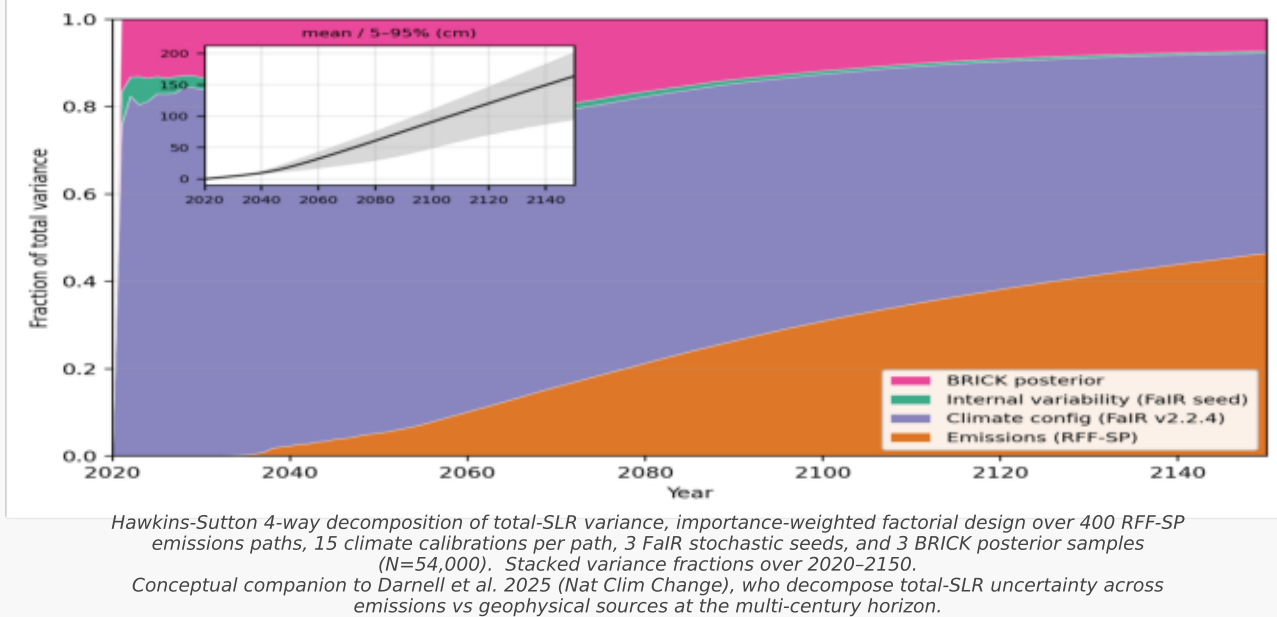
There is high value in probabilistic damage assessment for both mitigation-benefit estimates and adaptation-response planning. Here we demonstrate one methodology that leverages emissions uncertainty from RFF, climate uncertainty from FaIR, sea-level-response uncertainty from BRICK, global-to-local mapping from Sweet et al., and damage functions from FrEDI to produce probabilistic damage estimates by US state for three SLR-sensitive impact sectors: coastal property damage, high-tide-flooding transportation interruptions, and high-tide-flooding elder mortality. This pipeline allows fast estimation of damages for any future scenario.

Louisiana is the state most at threat for SLR-related impacts, particularly on a per-capita basis; Florida, Massachusetts, Virginia, and New Jersey are distant 2nd–5th in per-capita impacts (Panel I). Among FrEDI sectors, HTF elder mortality has the largest monetized impact (Panel J), with HTF transportation 2nd and coastal properties 3rd (Panel H). Climate-response uncertainty dominates total SLR variance in 2100, with emissions and BRICK uncertainty making secondary but meaningful contributions (Panel C); for the marginal pulse response, BRICK uncertainty dominates and AIS tipping makes internal variability a substantial contributor (Panel D).

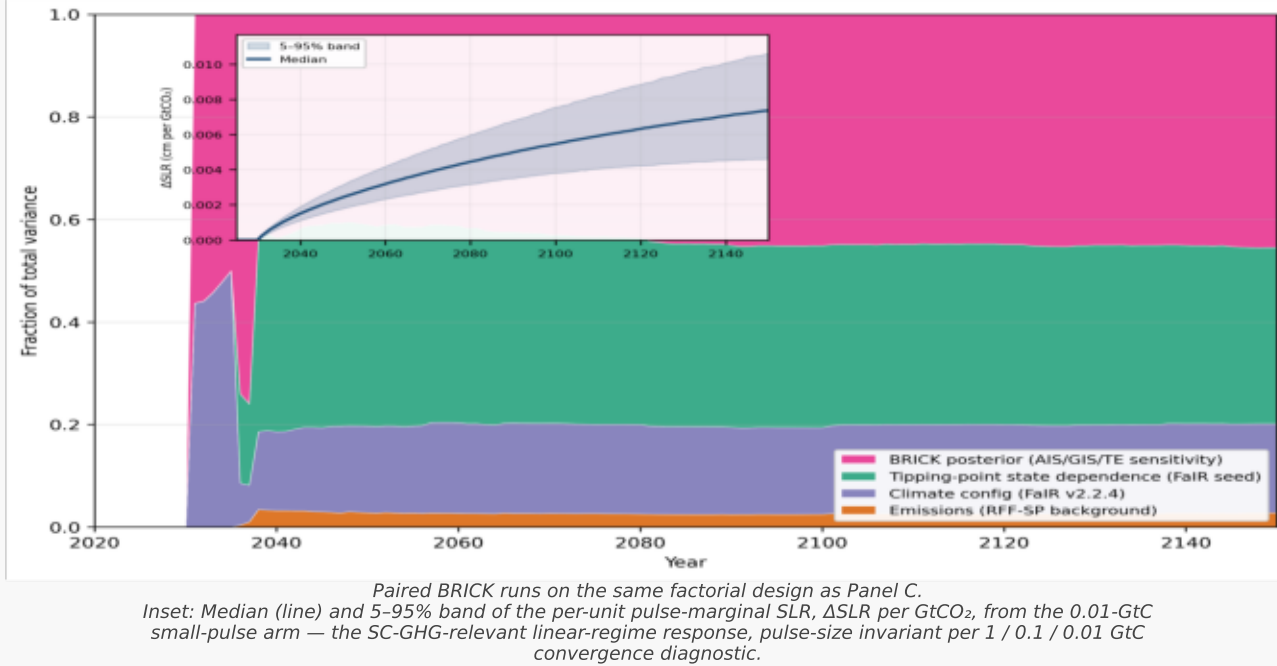
Key considerations we identify for this kind of work: dependence between FrEDI and BRICK parameter uncertainty (Wong et al. 2026); look-ahead-based adaptation estimates that account for observed non-optimal adaptive behavior and smoothing of capital expenditures (Panel G); and the importance of a wide range of anchor scenarios for damage-function calibration (Panel F).

The most important limitation we identify is the limited set of impact methodologies currently in FrEDI. We call on the community to produce more impact estimates that can be transformed into damage functions to inform this kind of analysis.

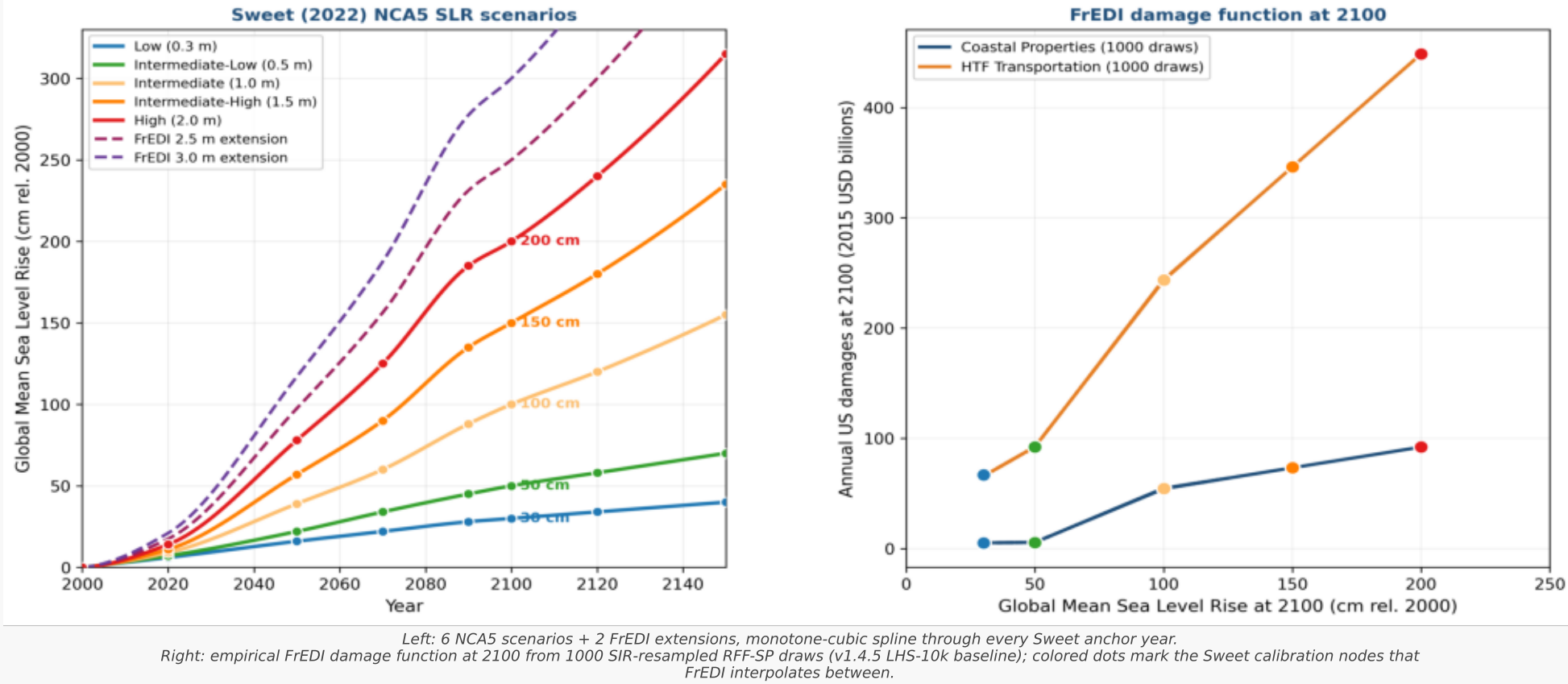
C. TOTAL SLR — sources of uncertainty



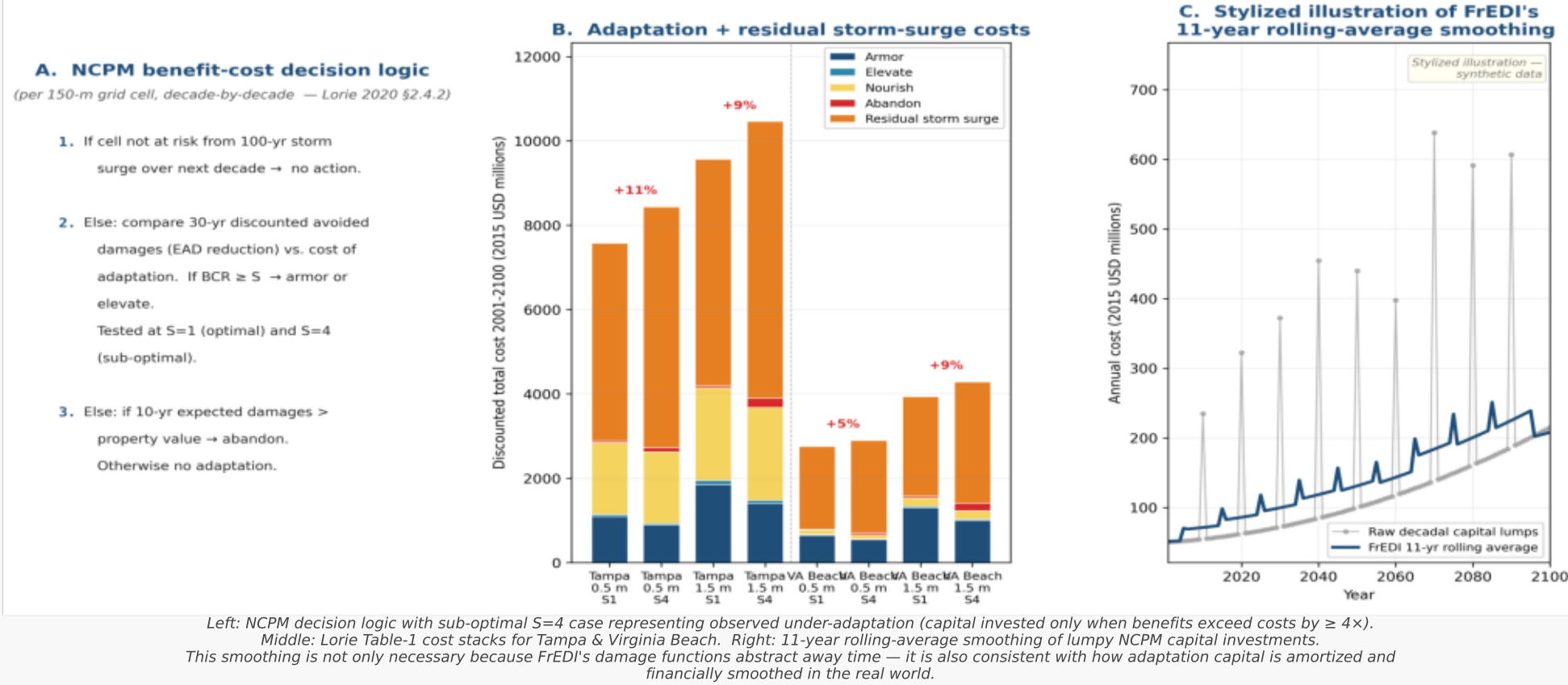
D. PULSE SLR — sources of uncertainty (1 GtCO₂ pulse at 2030)



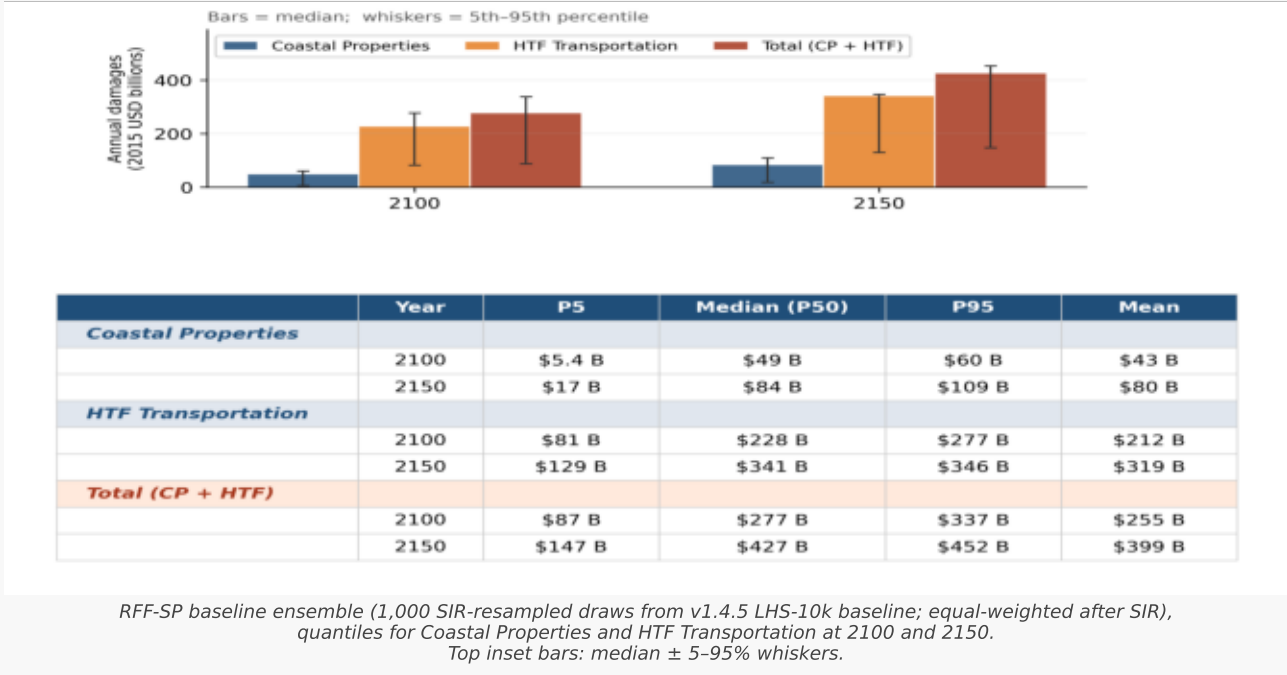
F. DAMAGE-FUNCTION METHODOLOGY — Sweet 6 SLR nodes + FrEDI damage curve



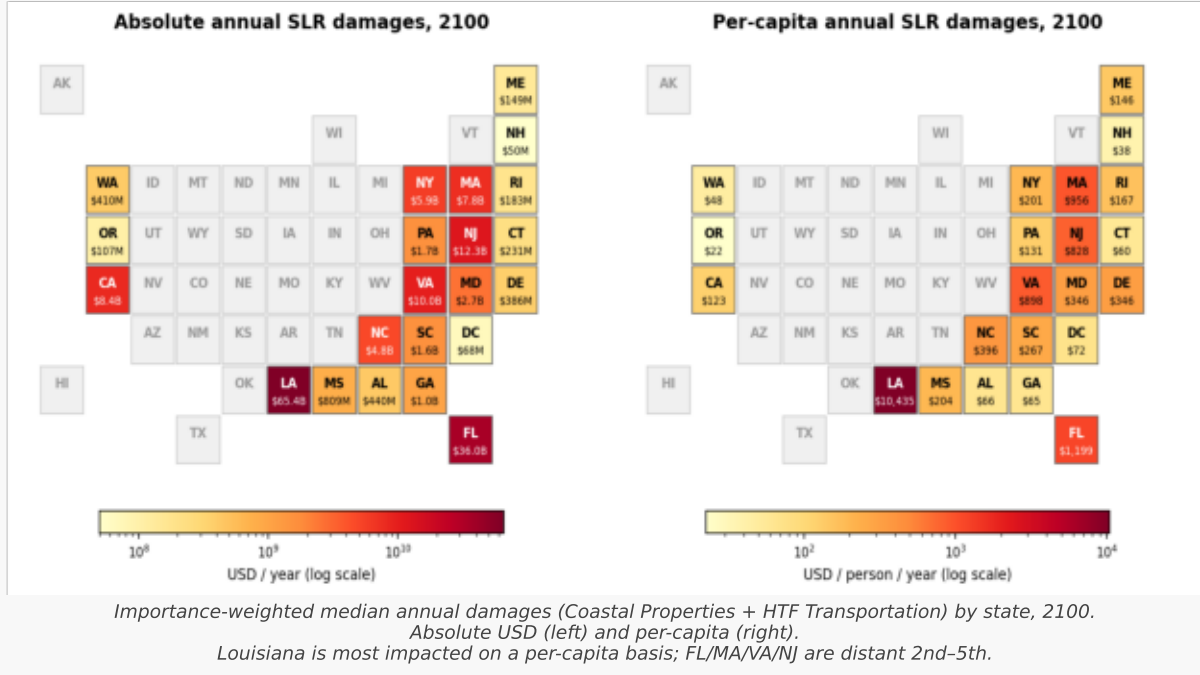
G. ADAPTATION (Lorie 2020) — NCPM decision logic + 11-yr smoothing



H. COASTAL PROPERTIES AND HTF TRANSPORTATION DAMAGES



I. COASTAL PROPERTIES AND HTF TRANSPORTATION DAMAGES BY STATE



J. HTF ELDER MORTALITY (Sheahan 2025)

	5th percentile	50th (median)	95th percentile
— No additional adaptation (Sheahan 2025 Table 2) —			
Additional deaths/yr, 2020	206	228	244
Additional deaths/yr, 2050	1,381	1,515	1,996
Additional deaths/yr, 2100	4,281	9,661	27,963
VSL-monetized damages, 2020 (USD billion)	\$2.8	\$3.1	\$3.4
VSL-monetized damages, 2050 (USD billion)	\$30.0	\$32.9	\$43.3
VSL-monetized damages, 2100 (USD billion)	\$170.9	\$385.6	\$1,116.0
— With stylized adaptation (cap at 95th-pct baseline) (Table 3) —			
Additional deaths/yr, 2100	2,765	5,555	13,312
VSL-monetized damages, 2100 (USD billion)	\$110.3	\$221.7	\$531.3
— Reduction at 2100 from stylized adaptation —	≈ 35%	≈ 43%	≈ 52%

Sheahan et al. 2025 Tables 2 & 3. Additional 65+ deaths / yr from high-tide flooding under the Rennert et al. (2022) RFF-FaIR-BRICK distribution. VSL = \$7.9M (1990s) inflated to 2023s.
Bottom row: % reduction from stylized adaptation at the 5th, 50th, and 95th percentiles.

K. CAVEATS

- RFF-SP emissions vintage: Scenarios were developed about 5 years ago; given rapid renewable / battery advances, present-day analogs might differ.
- FaIR known limitations: State-of-the-art probabilistic implementation, but missing permafrost feedbacks, possible Amazon dieback, and ozone / carbon-cycle interactions.
- BRICK upper bias: Runs slightly higher than the AR6 best estimates.
- Sweet et al. as a single realization: One realization of local SLR given global SLR; different partitioning between AIS, Greenland, thermal expansion, and local wind / current effects would yield different results.
- FrEDI sector coverage: Accounts for a limited number of SLR-derived damage categories; does not yet include probabilistic damages.

L. REFERENCES

- Darnell et al. 2025. Nat Clim Change. The interplay of future emissions and geophysical uncertainties for projections of sea-level rise. doi.org/10.1038/s41558-025-02457-0
- EPA. 2024. EPA 430-R-24-001. Technical Documentation for the Framework for Evaluating Damages and Impacts (FrEDI). U.S. Environmental Protection Agency.
- Fant et al. 2021. J Infrastructure Sys. Mere Nuisance or Growing Threat? The Physical and Economic Impact of High Tide Flooding on US Road Networks. [doi.org/10.1061/\(ASCE\)1519-3555\(2020\)100233](https://doi.org/10.1061/(ASCE)1519-3555(2020)100233)
- Lorie et al. 2020. Clim Risk Mgmt. Modeling Coastal Flood Risk and Adaptation Response under Future Climate Conditions. doi.org/10.1016/j.crm.2020.100233
- Neumann et al. 2021. Climatic Change. Climate effects on US infrastructure: the economics of adaptation for rail, roads, and coastal development. doi.org/10.1007/s10584-021-03179-w
- Rennert et al. 2022. Nature. Comprehensive evidence implies a higher social cost of CO₂. doi.org/10.1038/s41586-022-05224-9
- Sheahan et al. 2025. Lancet Planet Health. Projections of future mortality risk in older adults from high-tide flooding in coastal areas of the USA: an economic modelling study. doi.org/10.1016/j.lanphl.2025.101382
- Smith et al. 2024. Geosci Model Dev. fair-calibrate v1.4.1: calibration, constraining, and validation of the FaIR simple climate model for reliable future climate projections. doi.org/10.5194/gmd-17-8569-2024
- Sweet et al. 2022. NOAA Tech Rep NOS 01. Global and Regional Sea Level Rise Scenarios for the United States. earth.gov/sea-level-us/resources/2022-sea-level-rise-technical-report/
- Wong 2026. arXiv preprint. Modeling the Sea-Level Change from U.S. Vehicle Emissions. doi.org/10.48550/arXiv.2604.13446